

## 23.1 Requirements

### 23.1.1 Safety Information



#### CAUTION

The foamed plastic and protective film around the body resonator provides noise protection, do not remove or damage!

BC components can be easily damaged if not handled properly.

- » Never deposit the body resonator on a sharp-edged surface. Plastic parts and capacitors could be damaged. Use the delivered service tool for BC lifting. Use the shipping crate to store the BC.
- » To store the BC temporarily, always use the 2 foam blocks delivered with the replacement support tools.



#### CAUTION

During the BC replacement, you have to work close to the non-isolated connections of the gradient cables at the GC.

Hazardous voltages may be present at the gradient cable connections.

- » Switch OFF the gradient power amplifier via circuit breaker F21/EPC and F1 in the GPA. Wait at least 20 minutes before start working at the BC.



#### CAUTION

Heavy load! Weight of the BC is approximately 130 kg!

- » To lift the BC from and up to the patient table, 3-4 persons are required.
- » Use/mount the lifting handles at the BC.

### 23.1.2 Estimated Completion Time and Number of Persons



The Optical Length Meter (OLM) inside the BC is discontinued. Even, if still present at the old BC it should not be re-installed at the new BC.

Check / perform "Deactivation of OLM" in the following Update Instructions together with a BC replacement:

- MR005/15/P + MR019/15/R (Aera, Skyra)
- MR021/15/P (Skyra pTX)
- MR023/15/R (Aera 24Ch)

Schedule 30 min. additional time for the Update.

Tab. 38 Completion Time

| Work Steps            | Time                                                                                                                                              | Persons        |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Preparations          | 45 min.                                                                                                                                           | 1              |
| Removal/ Installation | 240 min.                                                                                                                                          | 2 <sup>1</sup> |
| Startup/ TuneUp/ QA   | 60 min.<br><br><b>Note:</b><br>additional<br>30 min. to<br>perform Up-<br>date In-<br>structions<br>to deacti-<br>vate OLM,<br>see note<br>above. | 1              |
| Final Steps           | n.a.                                                                                                                                              | n.a.           |
|                       | <b>Total = 5.5 hr.<br/>(6 hr. with UI)</b>                                                                                                        |                |

1. To lift the BC from and up to patient table, 3-4 persons are required. The weight of the BC is approximately 130 kg.

### 23.1.3 Aids or Tools

- Non-magnetic tool kit
- Non-magnetic measuring tape or gauge
- Waterproof marker
- Non-magnetic spirit level

- Non-magnetic torque wrench 5-20 Nm (material number 83 95 829)
- Accessory for torque wrench (material number 83 96 116)
- Electric screwdriver recommended (for opening/ closing the shipping crate)
- Non-magnetic flat tool (e.g. plastic ruler) to reattach the sound-absorbing foam pads.
- only if applicable: Sylomer parts Aera / Skyra / Skyra fit (material number 106 76 588)<sup>1</sup>



Only in case your system has an Optical Length Meter (OLM) installed which is not yet deactivated:

Check / order material for the following Update Instructions:

- MR005/15/P + MR019/15/R (Aera, Skyra)
- MR021/15/P (Skyra pTX)
- MR023/15/R (Aera 24Ch)

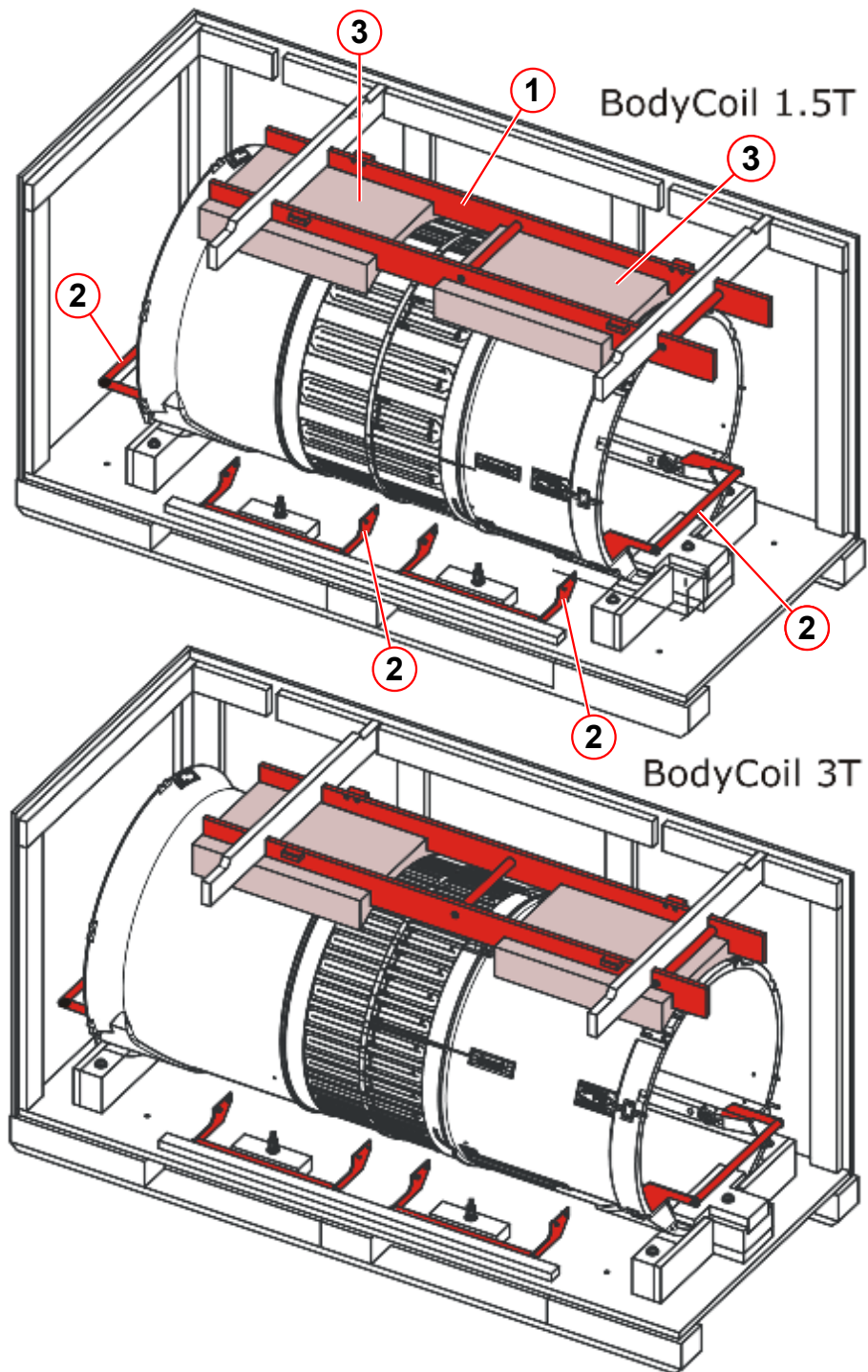
#### 23.1.3.1 Handling Tools delivered with the replacement BC

- Slide support for removing/ installing the Body Coil
- 2 additional lifting handles (+ 2 handles installed)
- 2 foam blocks for temporary BC storage at the ground

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1. (These parts are included in the new replacement BC. They must be separately ordered just in case the old Body Coil should be reinstalled but decoupling foam elements are damaged and need replacement)

Fig. 209: Replacement BC on pallet with tools

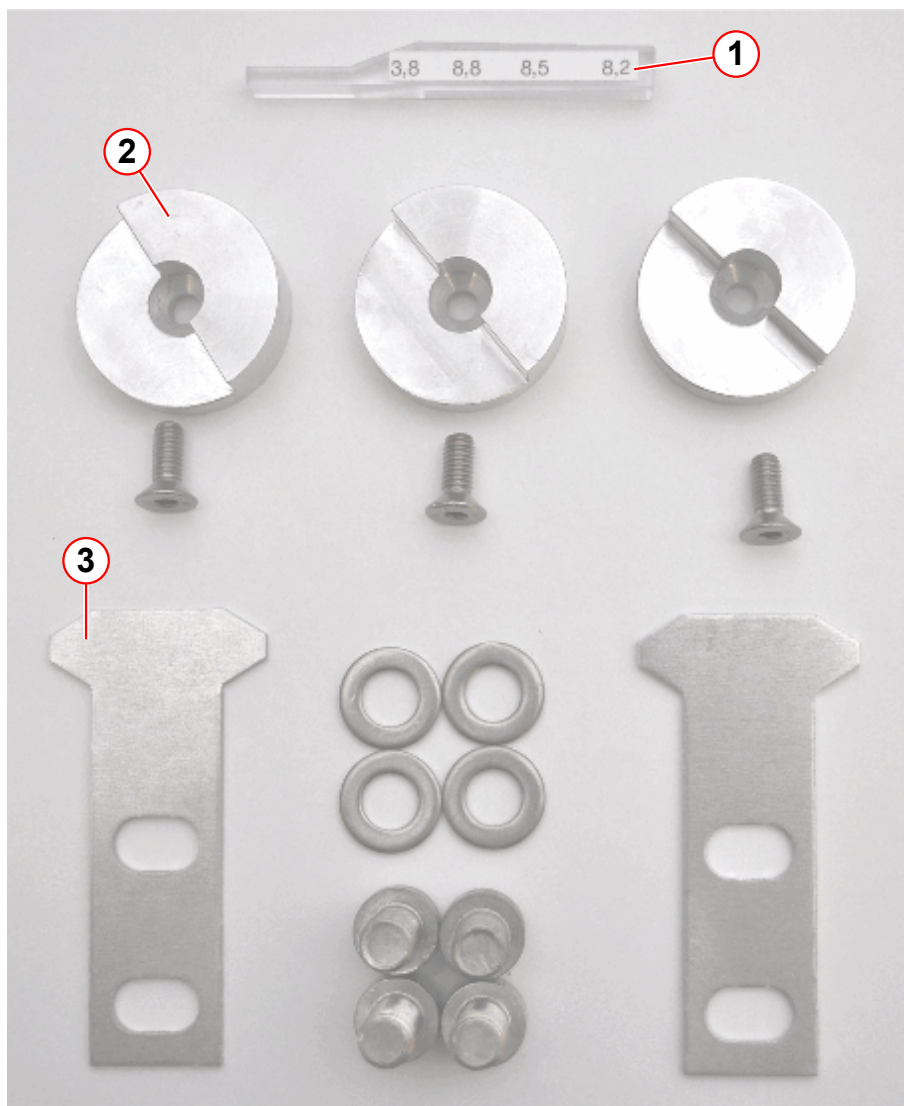


- (1) Slide Support
- (2) Lifting handles
- (3) Foam blocks for temporary BC storage

**Tool Set BC Replacement (part no.103 58 867) delivered with the spare part BC:**

- Feeler gauge GC-BC098
- 3 distance cylinders BC098
- 2 distance markers

Fig. 210: Additional Tools



- (1) Feeler gauge GC-BC098
- (2) Distance cylinder BC098
- (3) Distance marker



The tool set BC Replacement support Aera/Skyra (103 58 867) is contained in the replacement BC or can be ordered individually from the Service Parts Catalog (SPC).

#### Set of small parts, delivered with the spare part BC:

- Screws and caps
- Nuts
- Washers
- Decoupling foam pads (Sylomer parts Aera / Skyra / Skyra fit, material number 106 76 588)
- Table end stop, table rail covers, etc.

Fig. 211: BC small parts - example



## 23.2 Preparatory Work Steps

### 23.2.1 Magnet Cover parts and Power-off



For details of disassembling/assembling cover parts, refer to the Replacement of Parts, System.

Aera / Skyra, "General Information".

- Perform a QA Phantom Shim Check, to check the actual shim situation.  
If this is not possible, use the latest QA Phantom Shim report.  
This step is necessary to verify, there is no shim deviation after BC replacement.
- Remove the lower service cover at magnet.
- Move the patient table fully down.
- Remove the front funnel.
- Remove the front funnel segment.
- Select System/ Control/ Meas Recon/ Standby. Secure the system against switch-on.
- Switch off F21/EPC and the F1 circuit breaker in the GPA.
- After power-off, wait at least 20 min. to discharge GPA, then remove rear funnel.
- Remove the cover water hoses.

## 23.3 Removal

### 23.3.1 BC rail parts

**The end parts of the BC rails must be removed to attach the delivered handle bars:**

- Remove the plastic caps with a small flat screwdriver.

*Fig. 212: Removing plastic caps*



- Remove the rail end parts by releasing the attachment screws with a 5 mm Allen key.

*Fig. 213: Removing rail end parts*





- At older systems, remove the fixation pin with a 6 mm wrench.

Fig. 214: Removing the securing pin (only for older systems)



### 23.3.2 Optical Length Meter OLM (if applicable)



The Optical Length Meter (OLM) inside the BC is discontinued. Even, if still present at the old BC it should not be re-installed at the new BC.

Check / perform "Deactivation of OLM" in the following Update Instructions together with a BC replacement:

- MR005/15/P + MR019/15/R (Aera, Skyra)
- MR021/15/P (Skyra pTX)
- MR023/15/R (Aera 24Ch)

Schedule 30 min. additional time for the Update.

**In case an OLM is installed at the system it must be removed from the BC:**

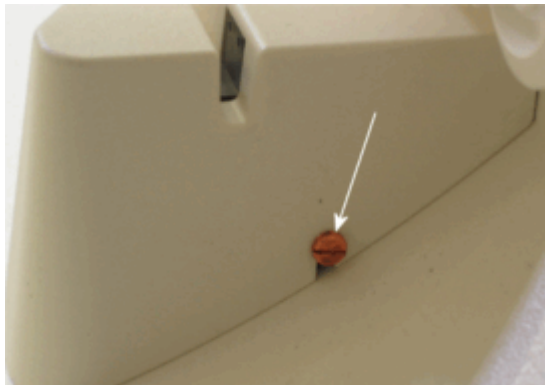
- Remove the plastic caps with a small flat screwdriver.
- Release the OLM housing by removing 3 screws (Allen 5 mm).

Fig. 215: Removing the OLM



- Loosen the screw in the back of the OLM housing.

*Fig. 216: Screw in back of the OLM housing*



- Disconnect and remove the OLM including FOCs from the BC.

*Fig. 217: OLM and FOCs*



The replacement BC is equipped with a modified cover and blind caps to cover the cavities of the eliminated OLM.

Fig. 218: Cover+blind caps for eliminated OLM - Aera

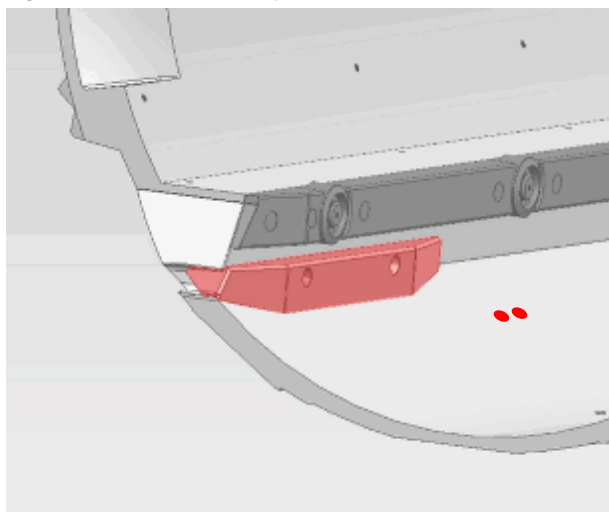
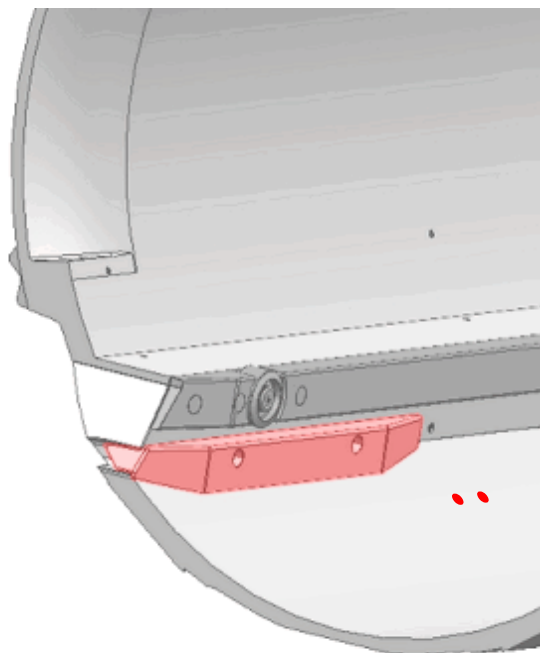


Fig. 219: Cover+blind caps for eliminated OLM - Skyra +Skyra fit

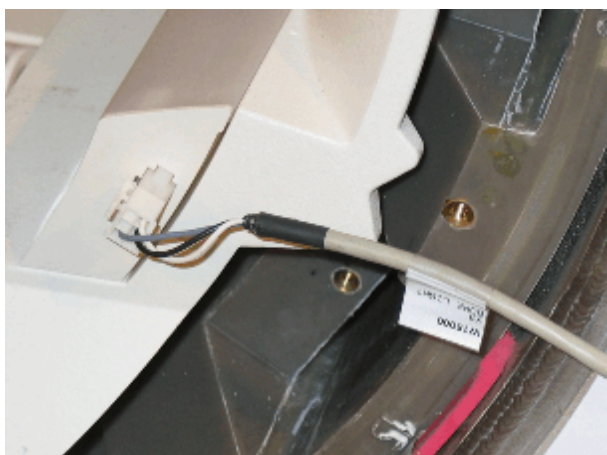


### 23.3.3 Cable Connections

**Disconnect the following cables at the front side of the BC:**

- Disconnect the 2 connectors to the bore lights on the front side (X3&X4) of the BC.

Fig. 220: Bore light connector front side



- Disconnect connector to the microphone board at the front side of the BC (at 11 o'clock position).

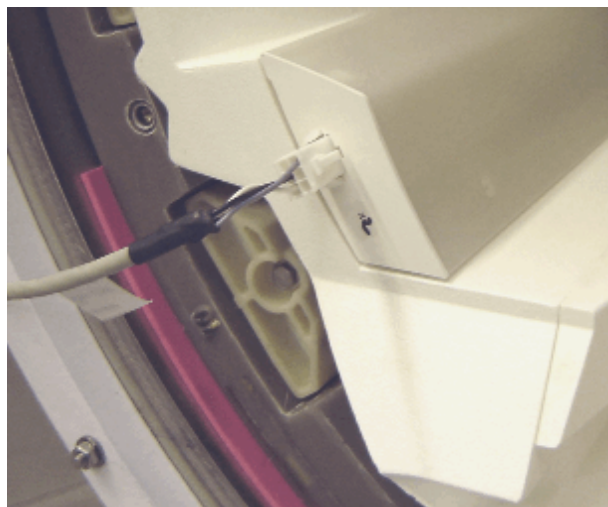
*Fig. 221: Microphone connector - front-side*



**Disconnect the following cables on the rear side of the BC:**

- Disconnect the 2 connectors to the two bore lights (X1&X2).

*Fig. 222: Bore light connector rear side*



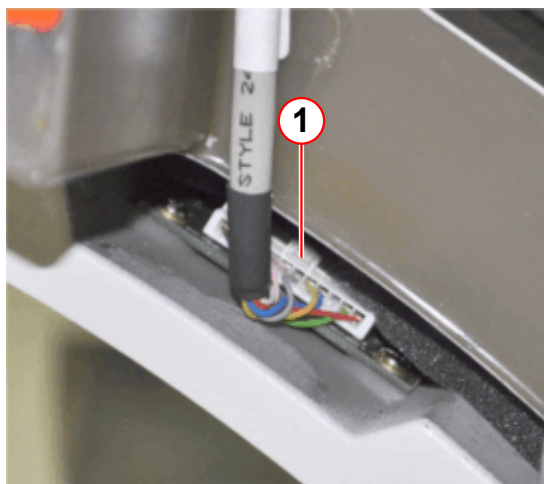
- At the BC rear-side, pull out the microphone board (at 11 o'clock position) from its mounting bracket with help of a small non-magnetic Allen key. Then, disconnect connector to the microphone board.

Fig. 223: Pulling out the microphone board at BC rear side



- Disconnect connector from the ID-Prom board (at 1 o'clock position). Use a small flat screwdriver to push down the latch (top Center Position) of the connector to release the lock mechanism.

Fig. 224: ID-Prom board with connector

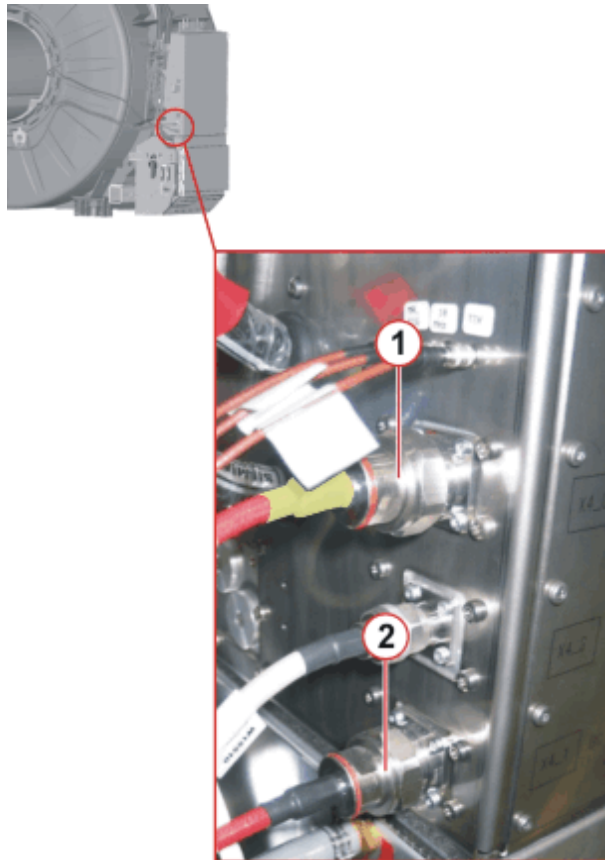


(1) Push down the latch to release the connector

**Disconnect the following cables of RF-components at the magnet:**

- Disconnect the 2 transmit cables at the TX-Box (BC\_0 and BC\_90).

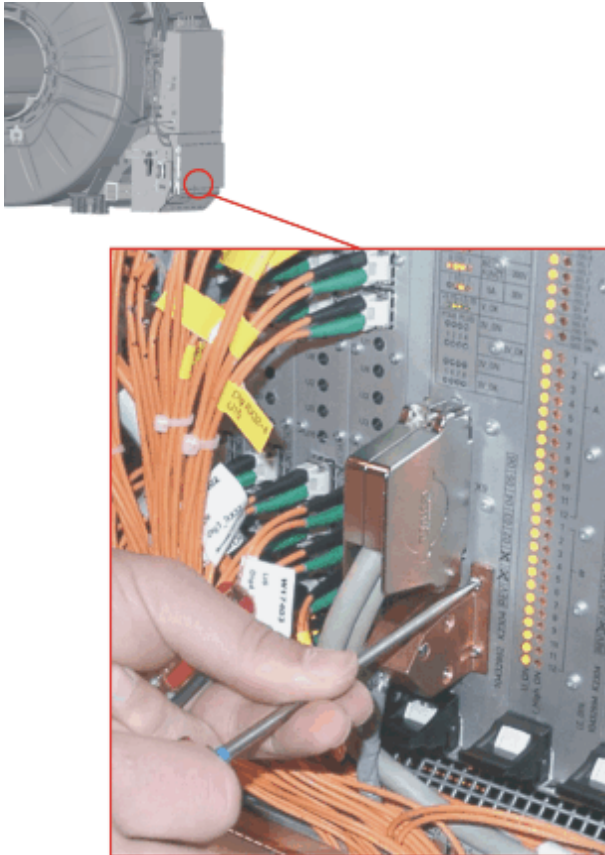
Fig. 225: Transmit cables at TX-Box - Example



- (1) TX-cable BC\_0
- (2) TX-cable BC\_90

- Loosen the 3 screws and remove the shielded PIN diode supply cable at X8 of the BC\_DYN in the RFCEL.

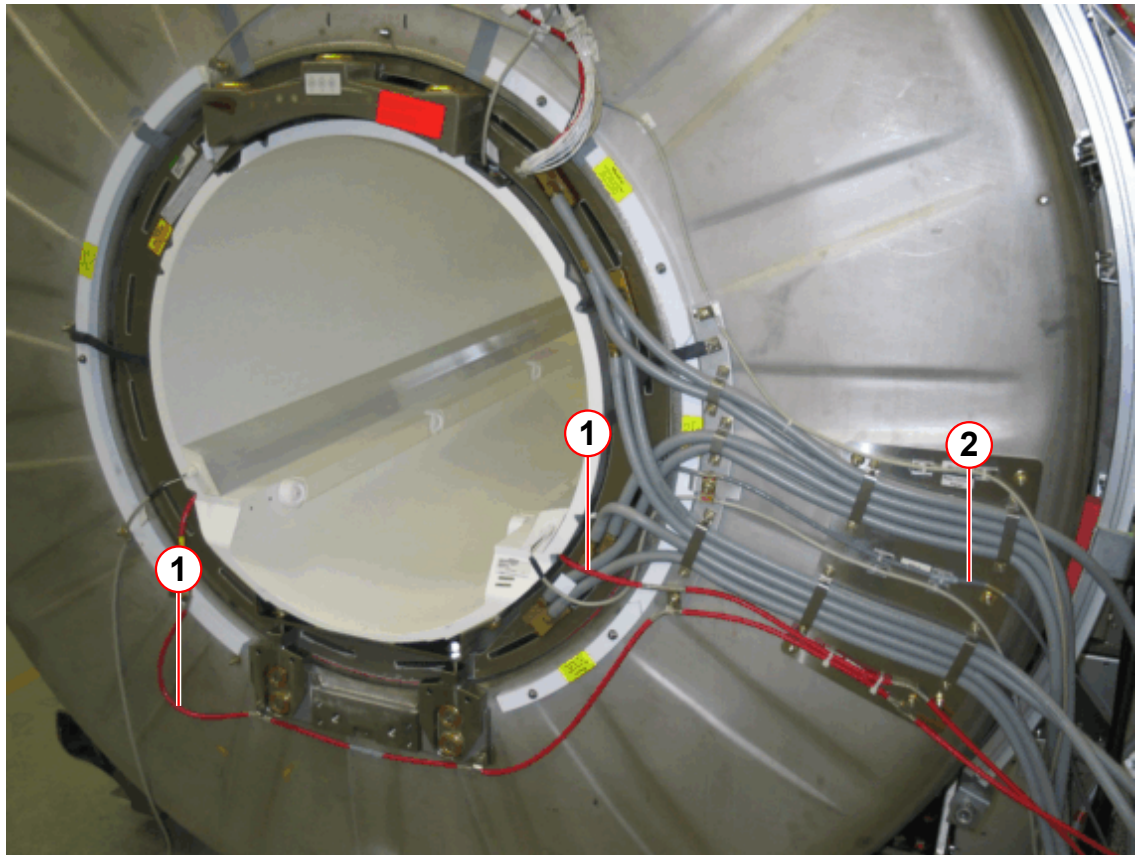
Fig. 226: Removing shielded connector X8 - BC\_DYN





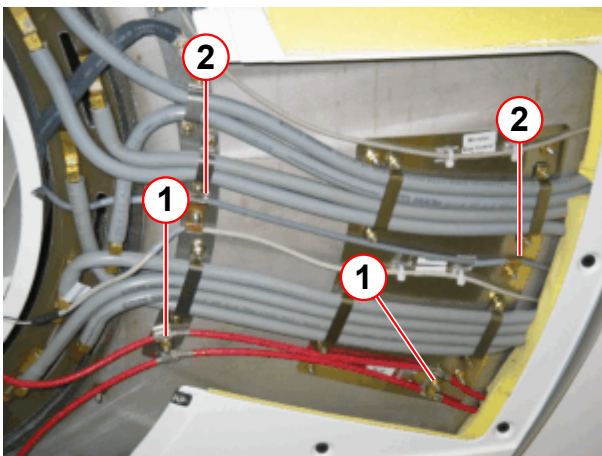
- Loosen, but do not remove the ground clamps of the 2 red RF cables (1).  
Loosen the gray PIN diode cable (2) at the magnet.

Fig. 227: RF-cabling in rear of magnet



- (1) RF-cables
- (2) PIN cable

Fig. 228: RF cabling



- (1) Ground attachments of red RF cables
- (2) Ground attachments of gray PIN diode cable



- Take the loose cables and roll them up to a loop. Then, place the cable bundle inside the BC tube and secure it with tape.

### 23.3.4 BC mechanical Removal

#### At the rear side of the old BC

- Remove the thin, black, sound-absorbing foam between BC and gradient coil.

Fig. 229: Foam removal



- In case of installed BC rear support, its **upper** part can be removed for better access to the BC screws.



#### CAUTION

Highly magnetic washers (labeled orange) are contained in the lower part of the rear support.

Risk of injury and component damage!

- » Never remove the lower part of the rear support under magnetic field!

#### NOTICE

Attention, the upper part of the rear support contains slightly magnetic parts !

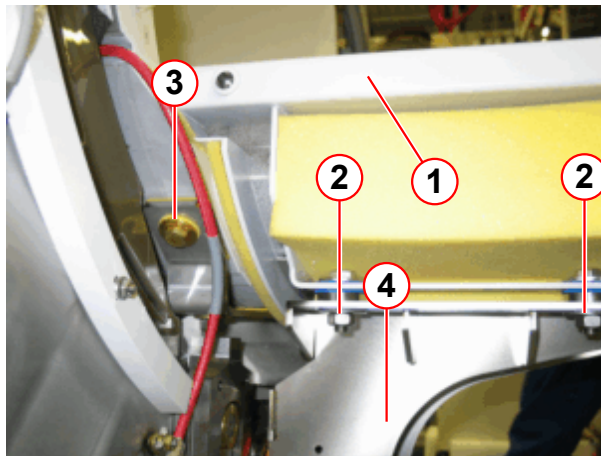
Risk of minor component damage if parts are slightly attracted by the magnetic field.

- » Hold slightly magnetic parts firmly with the hand and remove it immediately out of the magnetic field!

- Mark the z-position of the upper part of the rear support relatively to the lower part at left and right side with a waterproof marker pen.

- Loosen 4 nuts (2) (2 at left- and right-side) at the lower side of the upper part of rear support and remove it.

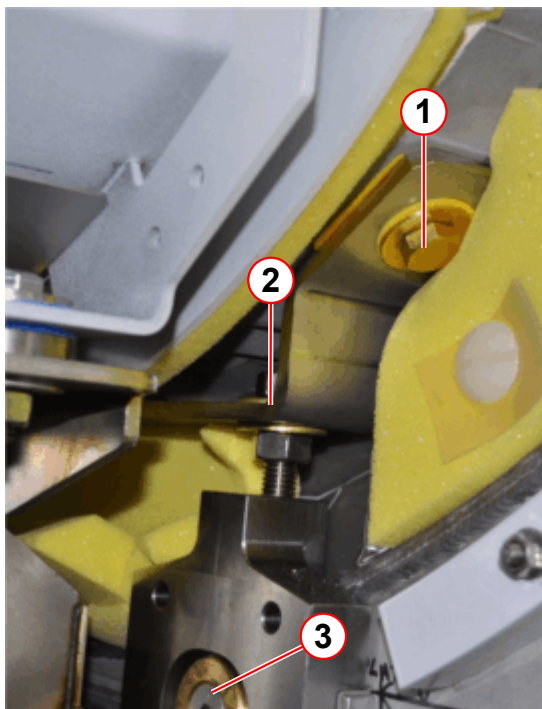
Fig. 230: Removal of upper part of BC rear support



- (1) Upper part of BC rear support
- (2) Remove nuts
- (3) BC mounting screws (to be removed)
- (4) Lower part of BC rear support

- Remove just the 2 BC screws (1), attaching the coil to the U-formed support bracket.
- The BC support bracket itself, including nuts (2) and Allen screws (3), remain installed.

Fig. 231: BC support in rear (with installed rear support)

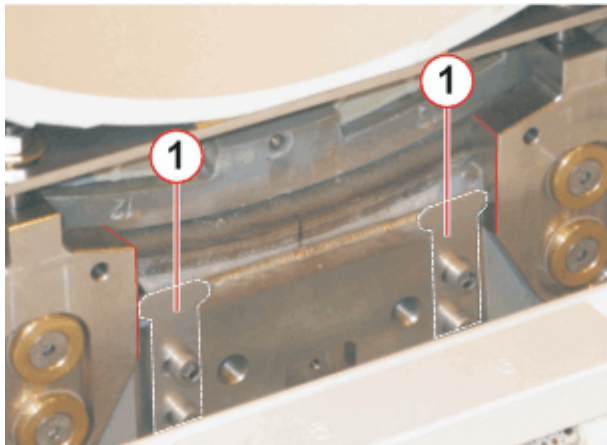


- (1) BC mounting screw (to be removed)
- (2) Adjustment nut and clamping nut (remain fixed)
- (3) Allen screw (6mm) at support (remains installed)

**At the front side of the old BC**

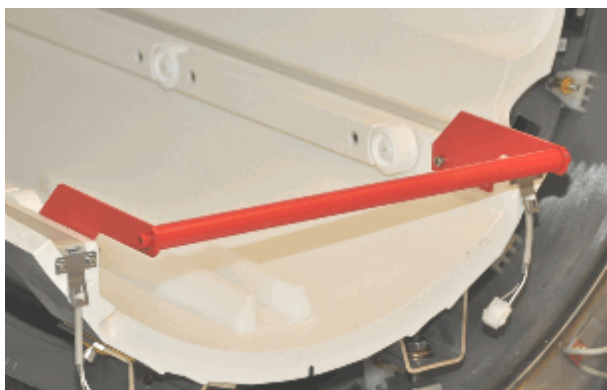
- Mount the 2 X-position markers with the delivered screws to the magnet (to capture the horizontal x-position reference for later BC reinstallation).

Fig. 232: Mounted X-position markers



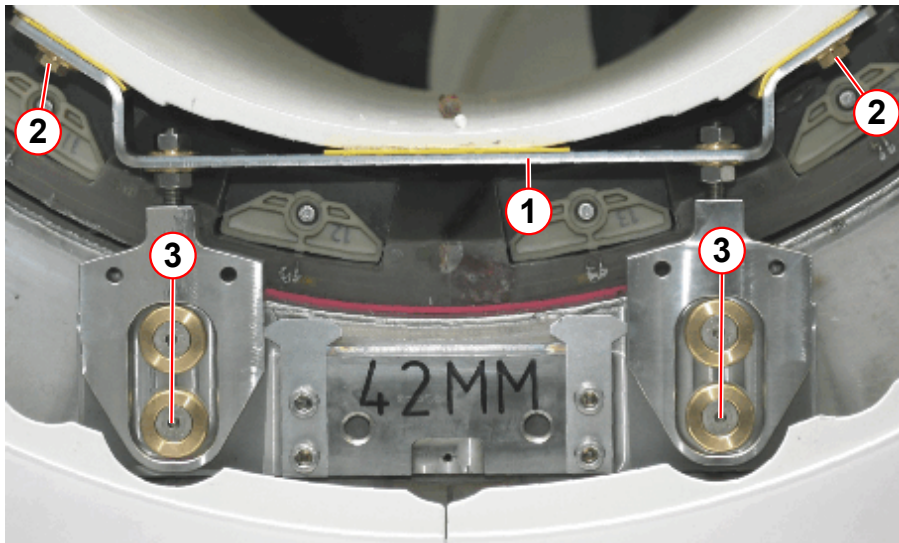
- Mount the 2 additionally delivered handles to the front and rear of the BC.

Fig. 233: Mounted BC-handle



- Disassemble the U-shaped BC support (1) completely by removing the 2 BC screws (2) and the 4 BC support screws (3) at the magnet.

Fig. 234: BC front support



- (1) BC support
- (2) BC mounting screw
- (3) BC support screw



To access the BC support screws, the patient table must be fully lifted up or undocked.



Do not manipulate the BC height adjustment nuts. They remain fixed to ensure the correct reassembling height for the new BC.

- Position the slide support on the patient table and adjust its height with reference to the BC.

Fig. 235: Slide support on patient table



- (1) Slide support

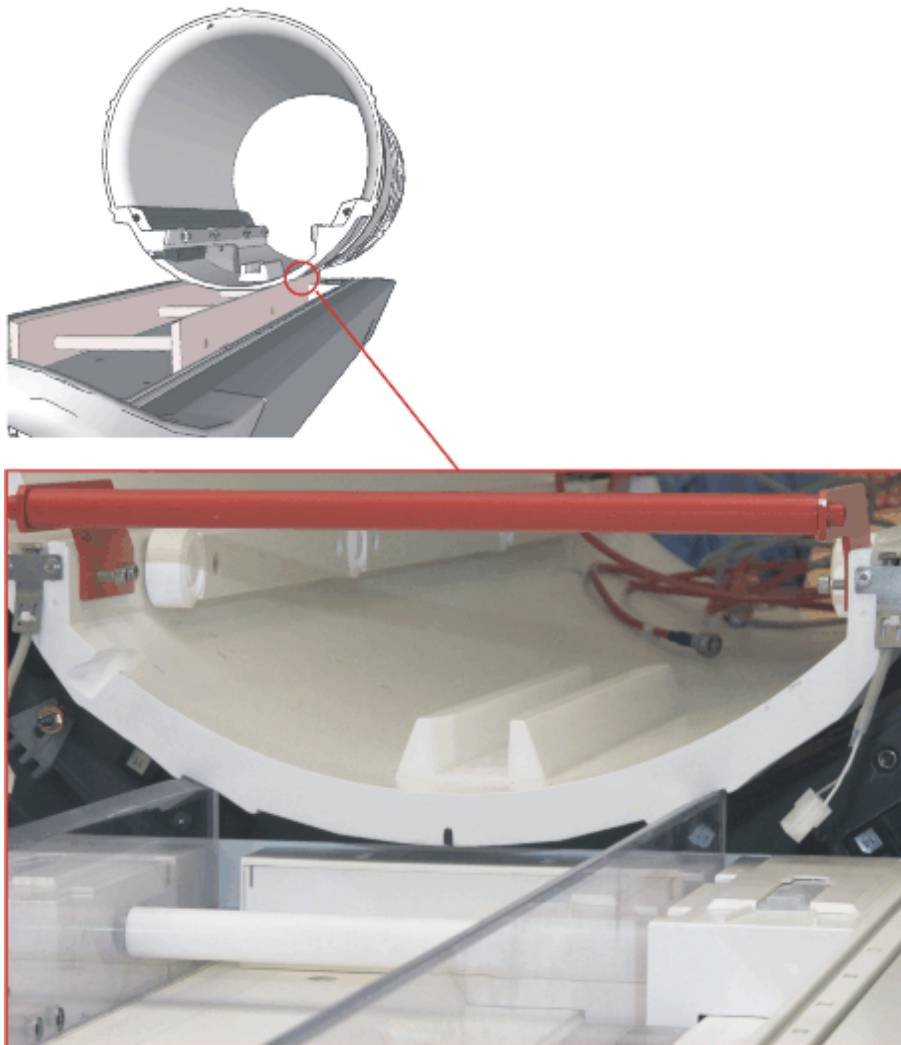
#### Adjusting the vertical table height for BC removal:

- For a mobile table, use the foot pedals.

- For the fixed table, engage the manual pump with the service lever (stored under the service access door of the foot cover).

The height of the slide support rails must align with the BC support guide rails.

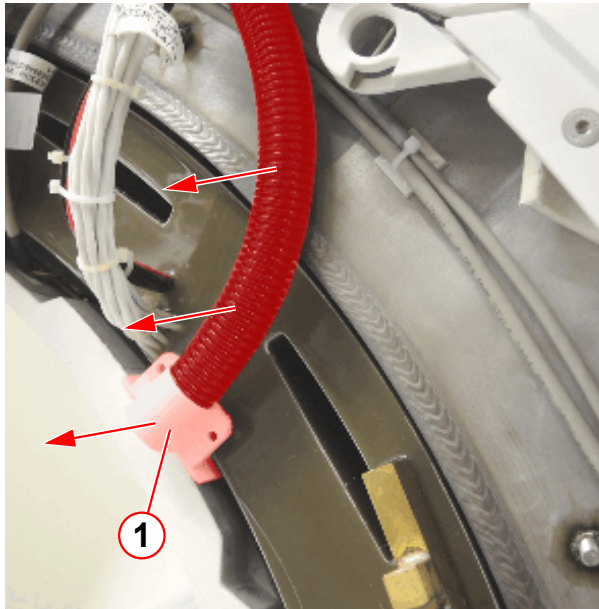
*Fig. 236: BC support guide rails*



**Remove the old BC:**

- If your system has a VLM aspirating nozzle at the BC, carefully pull out the nozzle assembly from the BC mounting bracket.

Fig. 237: VLM aspirating nozzle at BC - Example



(1) Pull out nozzle assembly

**NOTICE**

The hose is glued to the nozzle assembly.

» Pull out the complete nozzle assembly. Do not separate the hose!



2 persons are required for the following working steps.



- At magnet rear-side, lift the BC a little bit from its support and carefully push the BC approximately 5 to 10 cm towards magnet front side.

Fig. 238: Lift up and move BC



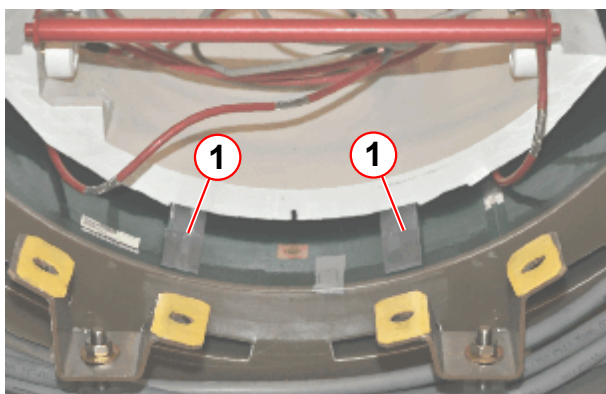
### ⚠ CAUTION

When removing the BC from its supports, ensure that you don't jam your hands.

- » Use protective gloves. Slowly move the BC and latch its studs to the GC slide rails.

- Carefully lift-off the BC (rear-side) from the mounting support brackets, push it further to magnet front-side and make sure it is properly engaged to the GC slide rails.

Fig. 239: BC on slide rails in the GC



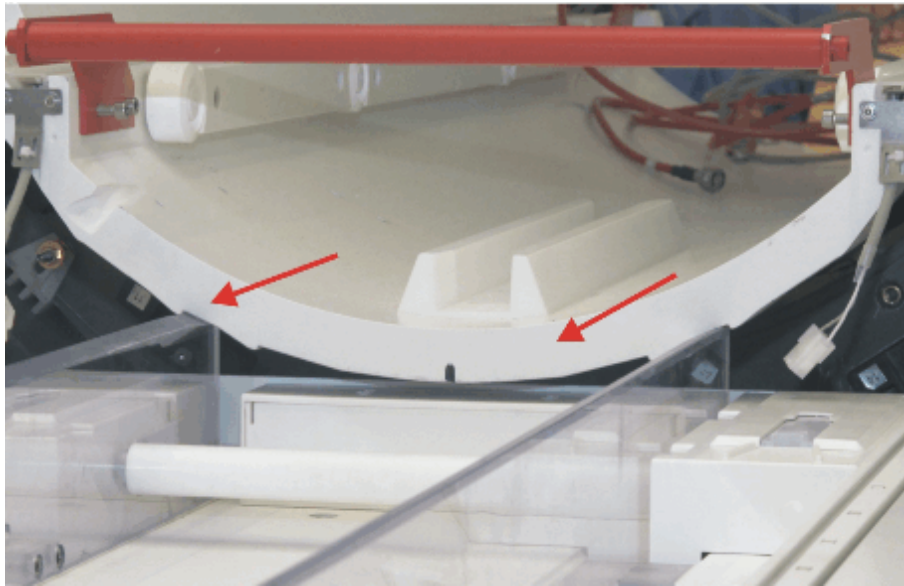
(1) GC slide rails

- Slowly pull out the BC from the GC towards magnet front side. The guide studs of the BC must latch accurately into the support guide rails.



- If necessary readjust the patient table vertical height for a smooth transfer.

Fig. 240: Pull out BC from GC

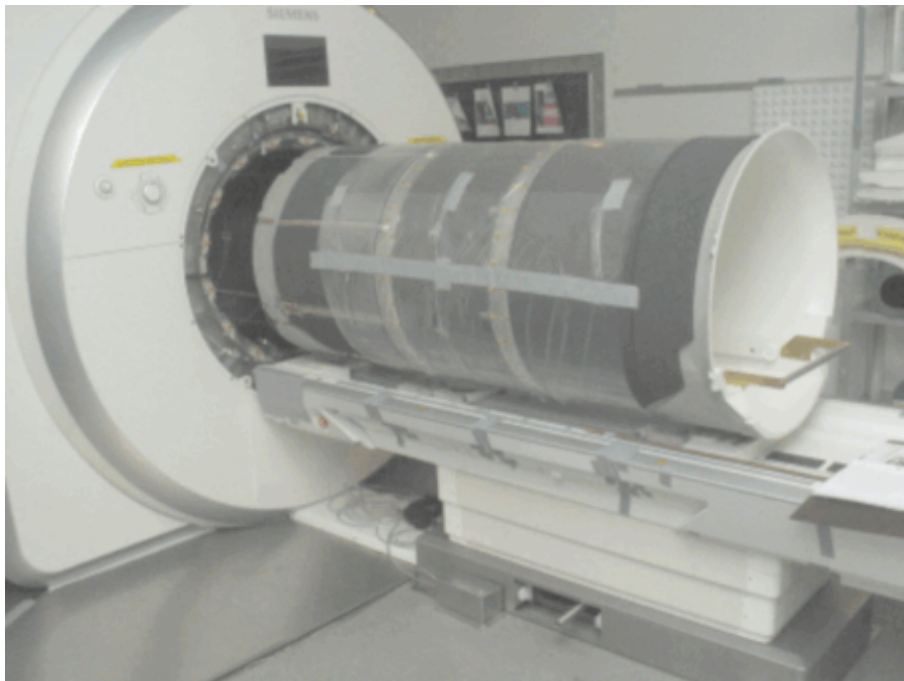


Use extreme care when removing the old BC to avoid damage to the Faraday Shield inside the gradient coil.

Carefully monitor the proper BC centering inside the GC.

- Lift off the BC (3-4 persons) from the patient table.

Fig. 241: BC pulled out from gradient coil on the slide support



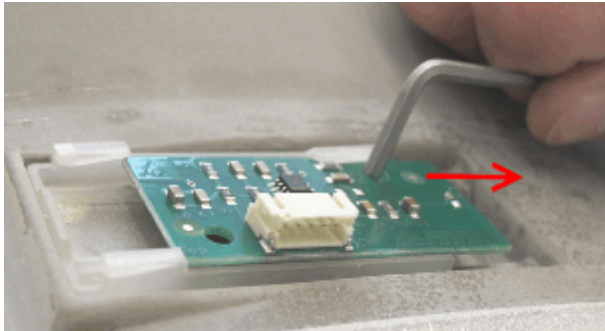
- Temporary store the old BC on top of 2 foam support cushions delivered with the BC.

### 23.3.5 Exchange of Attachments in the BC Bore

#### Microphone board front-side

- Use a small, non-magnetic Allen key, insert the short end in the hole of the microphone board and pull the PCB out of the clamping bracket.

Fig. 242: Removal of microphone board



- Attach the microphone board to the new BC in reverse order.

#### Microphone board rear-side

- Re-attach the rear microphone board of the old BC to the new BC in the same fashion as before.

#### Bore lights

- Remove left and right bore lights by removing 5 plastic screws each (flat screwdriver).

Fig. 243: Removing the bore light



Plastic screws are fragile, do not overturn.

- Remove the 2 attachment rails on each side of the BC and mount the rails to the new Body Coil (flat screwdriver).

*Fig. 244: Removing the bore light mounting rails*

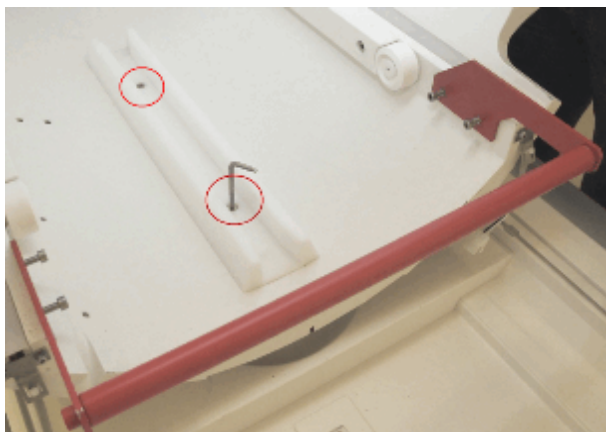


- Reattach the two bore lights to the new BC

### Chain Guide

- Remove the chain guide from the old BC (two 4 mm Allen screws) and re-attach it to the new Body Coil.

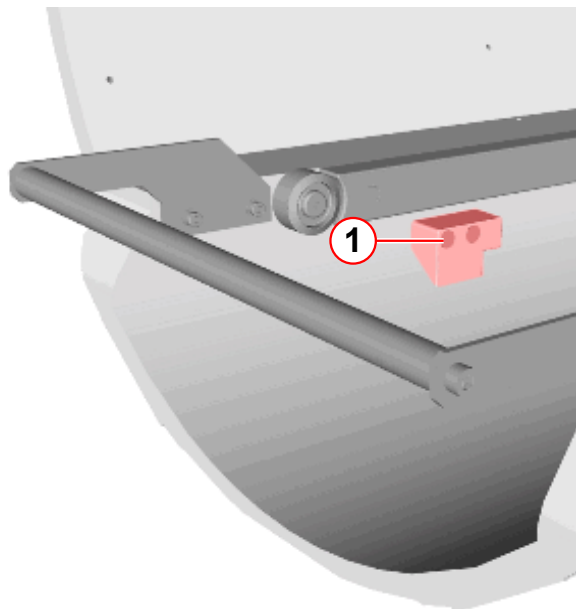
*Fig. 245: Assembling the Chain Guide*



**Table End Stop**

- Remove the table end stop from the old BC (two 4 mm Allen screws) and re-attach it to the new Body Coil.

Fig. 246: Body Coil - service end



(1) Table End Stop



For systems equipped with the Whole Body option, the table end stop is mounted at the rear-support.

## 23.4 Installation

### 23.4.1 BC transfer into bore

- Place the new BC on the slide support rails at the patient table using the lifting handles. The guide studs of the BC must accurately latch in the rails.
- Move the BC forward to the gradient coil and adjust the height of the patient table. The height of the slide support rails must align with GC slide rails.



Use extreme care when removing the old BC to avoid damage to the Faraday Shield inside the gradient coil.

Carefully monitor the proper BC centering inside the GC.



#### CAUTION

When placing the BC on the supports, ensure that you don't jam your hands.

- » Use protective gloves. Slowly move the BC on its studs at the GC slide rails.



3 to 4 persons are required for the following working steps.

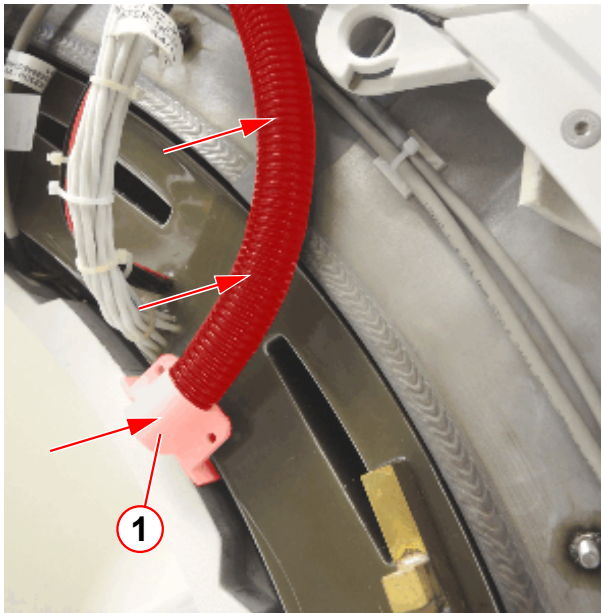
- Slide the BC carefully into the gradient coil.  
A second person at magnet rear-side must monitor the smooth transfer to avoid mechanical damage to the BC and gradient coil.
- Stop approximately 10 cm before the final position and lift the BC carefully onto the rear mounting support brackets. Secure the BC with its 4 screws (just hand-tight).
- At the front, lift the BC approximately 20 mm to have enough clearance for mounting the complete BC-supports underneath. For lifting, use the provided BC-handles.

Fig. 247: Mounting the BC supports



- Fix the mounting supports with four 6 mm Allen screws at magnet front-side. The previously installed X-position markers, see (→ Fig. 232 Page 306), will help you to re-establish the correct position adjustment.
- At magnet front- and rear-side, fasten the BC to the U-formed support bracket with 2 screws each (hand-tight).
- If your system has a VLM aspirating nozzle at the BC, carefully push the nozzle assembly into the BC mounting bracket.

Fig. 248: VLM aspirating nozzle at BC - Example



(1) Push-in nozzle assembly



2 persons are required for the following working steps.



### CAUTION

When removing the BC from its supports, ensure that you don't jam your hands.

- » Use protective gloves. Slowly move the BC and latch its studs to the GC slide rails.

#### Carefully check all Sylodyn BC decoupling foam parts:

- Carefully check that all Sylodyn foam parts (→ Fig. 249 Page 318) are undamaged and are mounted in a correct fashion.



- Replace damaged Sylodyn foam with parts contained in the kit.

#### NOTICE

##### **Risk of damaging Sylodyn BC decoupling foam parts!**

If Sylodyn parts are damaged or incorrectly assembled, the BC mechanical decoupling is deteriorated. This may lead to image quality problems like signal voids at diffusion weighted images (DTI) or spikes!

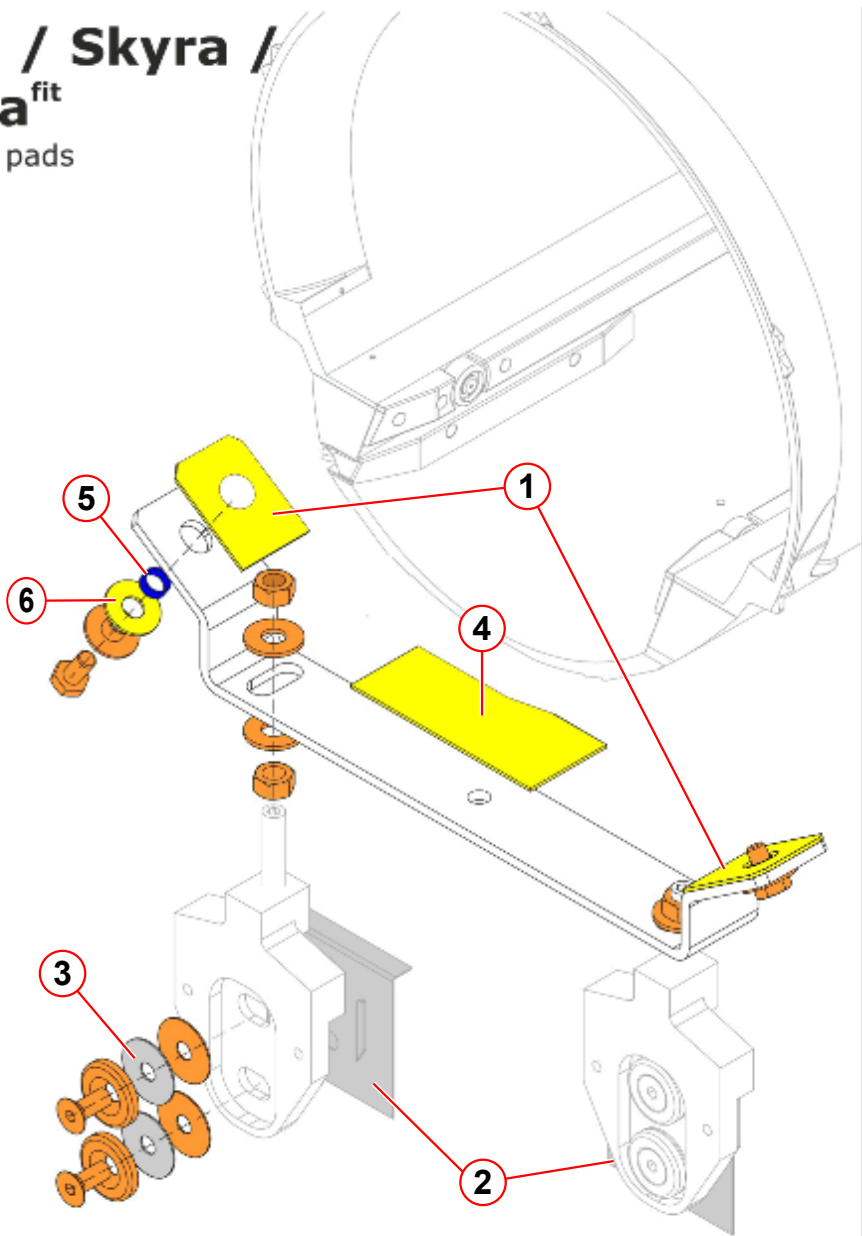
- » Ensure all necessary Sylodyn foam formed parts are undamaged and mounted in a correct fashion.



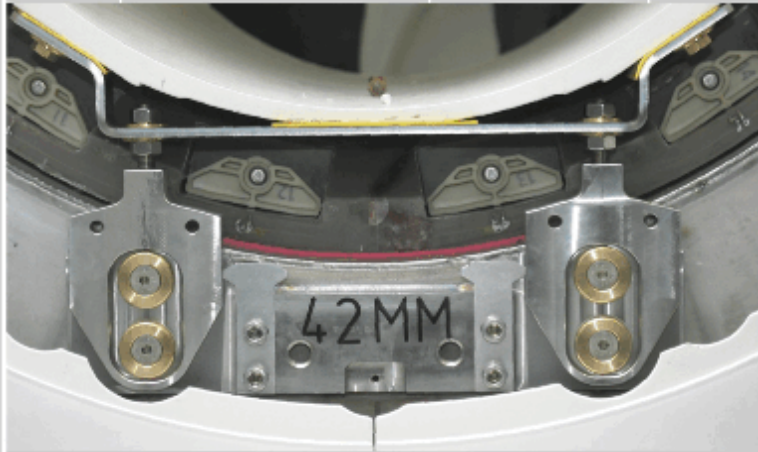
The Sylomer parts Aera/Skyra/Skyra fit are contained in the new replacement BC or can be separately ordered from the SPC (material number 106 76 588).

Fig. 249: BC decoupling foam parts

# Aera / Skyra / Skyra<sup>fit</sup> BC foam pads



| Pos.:    | Part:                                                           | Material no.:   | Qty.: |
|----------|-----------------------------------------------------------------|-----------------|-------|
|          | <b>Sylomer parts<br/>Aera / Skyra / Skyra fit<br/>contains:</b> | <b>10676588</b> |       |
| <b>1</b> | Support/rest                                                    | 10358329        | 4x    |
| <b>2</b> | Pad/bracket                                                     | 10048151        | 4x    |
| <b>3</b> | Sylomer disc                                                    | 10048153        | 8x    |
| <b>4</b> | Damping pad/press slice                                         | 10358844        | 2x    |
| <b>5</b> | Hollow cylindra<br>15x12.5x6                                    | 10358330        | 4x    |
| <b>6</b> | Disc D31 d12.6                                                  | 8121027         | 4x    |
|          | Adhesive pad                                                    | 10676497        | 3x    |



### 23.4.2 Mechanical BC Adjustment

- Mount the 3 distance cylinders at the 4, 8 and 12 o'clock position to the BC.

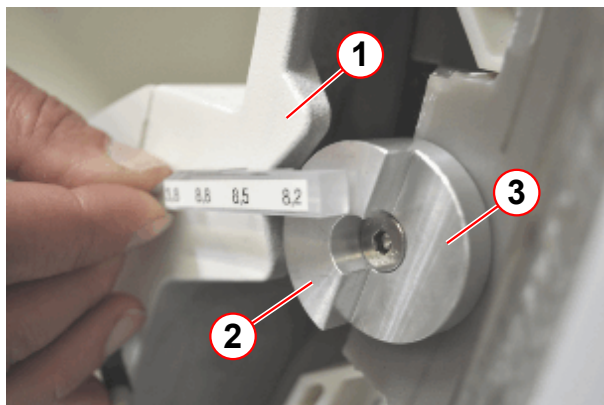
Fig. 250: Mounted distance cylinder at gradient coil (aligned for 1.5T systems)



- **Adjust the BC in the Z-direction:**

The distance cylinders can be rotated to set the different BC z-position adjustment length required for 3T or 1.5T systems.

Fig. 251: Front surface BC



- (1) Front surface BC
- (2) Z-distance alignment for 3T
- (3) Z-distance alignment for 1.5T

Fig. 252: Aligning the z-distance of the BC (for 1.5T)



- The adjustment is correct if the BC-front surface aligns with the corresponding top of the distance cylinder:
  - Distance for 3T systems: 16 mm to gradient coil edge surface.
  - Distance for 1.5 T systems: 8 mm to gradient coil edge surface.
- **Adjustment in the X- and Y-direction:**

The Body Coil has to be radially centered at both ends.

The distance between the Body Coil and the gradient coil must be equal all around the whole circumference (tolerance +/- 1 mm).

#### At the front side

Measure the 2 clearances between BC and each distance cylinder (mounted at 4, 8 and 12 o'clock position). Check the distance with the small end of the BC feeler gauge and ensure all clearances are equal.

Fig. 253: Checking the distance on the upper side



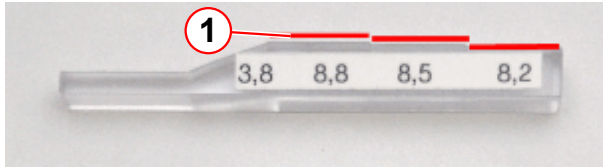
Fig. 254: Checking the distance at the lower side



**At the rear side**

Measure at 4, 8 and 12 o'clock position using the BC feeler gauge. Insert the gauge into the gap between BC and GC.

Fig. 255: Feeler gauge GC-BC098



(1) Distance measurement steps

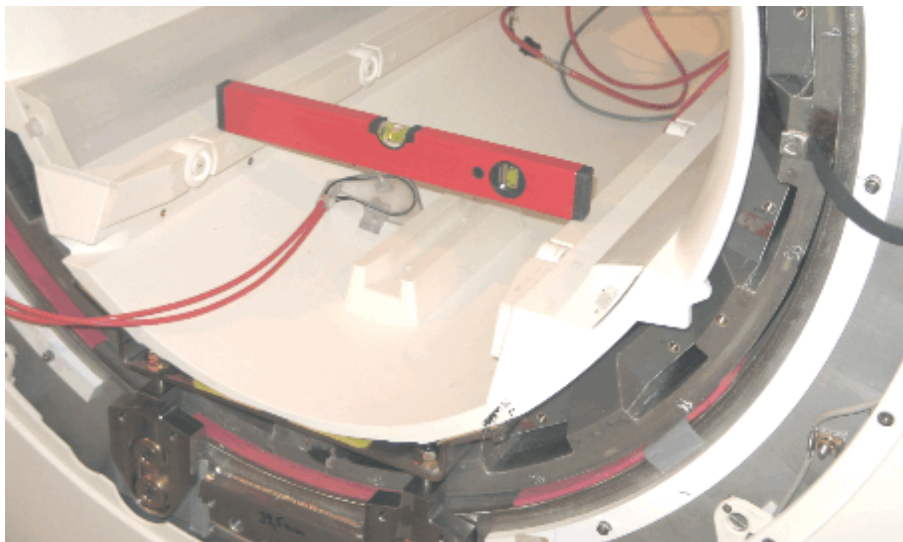
Fig. 256: Measuring the radial distance between BC and GC



For measurement, the feeler gauge side with steps must point towards the inner diameter of the gradient coil.

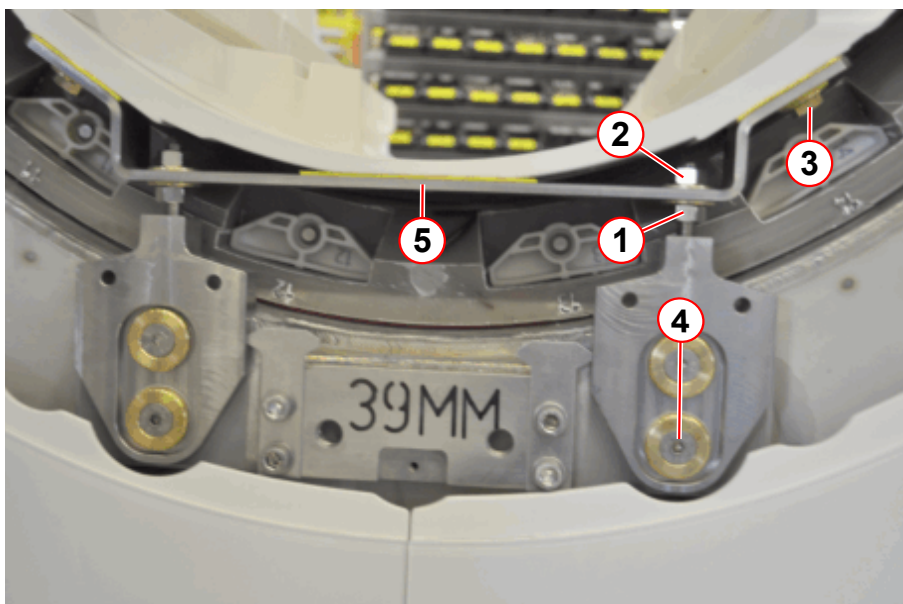
- Use a non-magnetic spirit level to align the Body Coil horizontally in the X-direction.

Fig. 257: Aligning the X-direction with a spirit level



- Adjust BC using the **adjustment nuts (1)** at the support brackets.

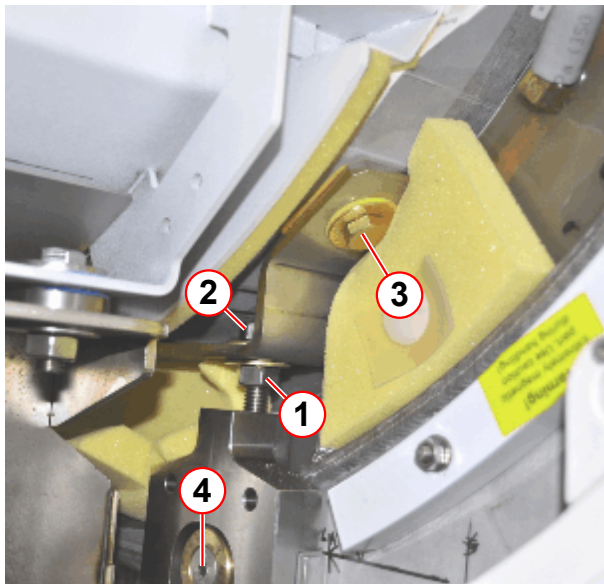
Fig. 258: BC support - front-side



- (1) Adjustment nut
- (2) Clamping nut
- (3) BC mounting screw
- (4) Allen screw at support
- (5) U-shaped mounting support bracket

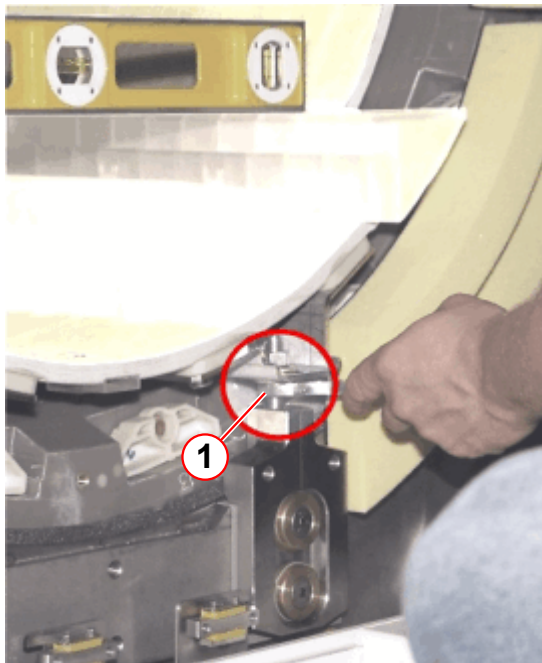


Fig. 259: BC support - rear-side



- (1) Adjustment nut
- (2) Clamping nut
- (3) BC mounting screw
- (4) Allen screw at support

Fig. 260: Adjusting BC using adjustment nut at BC support



- (1) Adjustment nut

- **If all measured distances are in specification:**
  - Tighten all screws and nuts of the coil supports:
  - Tighten the BC screws.

- Check the distances again; if not ok, repeat the adjustments

### 23.4.3 Installation of BC Cables and grounding

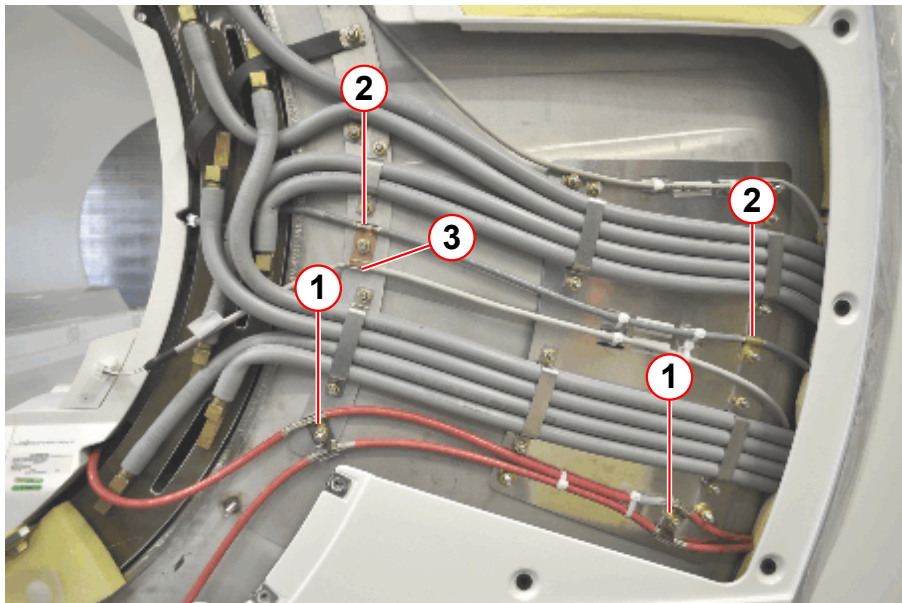
- Dismount the lifting handles.
- Install the removed upper part of the rear support (if existing):

**Attention weak magnetic object !**

The clearance between rear support and BC should be 4 mm to allow proper attachment of the seal.

- Route the RF cables from the BC back to the TX-Box.

Fig. 261: Cabling magnet rear - Grounding points

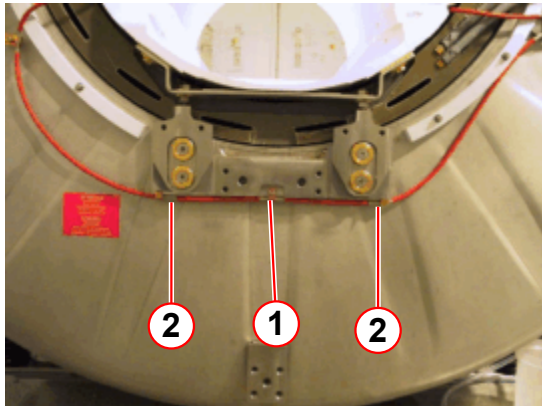


- (1) Grounding points RF cables
- (2) Grounding points- Pin diode cable
- (3) Grounding point - bore light

- Route and connect the PIN diode cable back from BC to the RFCEL/ BC\_DYN X8.
- Re-connect the cables to the bore lights, light marker, both microphones and the EEPROM board of the BC. Make sure the cables are properly routed and fix them with cable ties if necessary.

- Secure the RF cables carefully with clamps at the defined ground points of the magnet.

Fig. 262: Grounding point with/without rear support

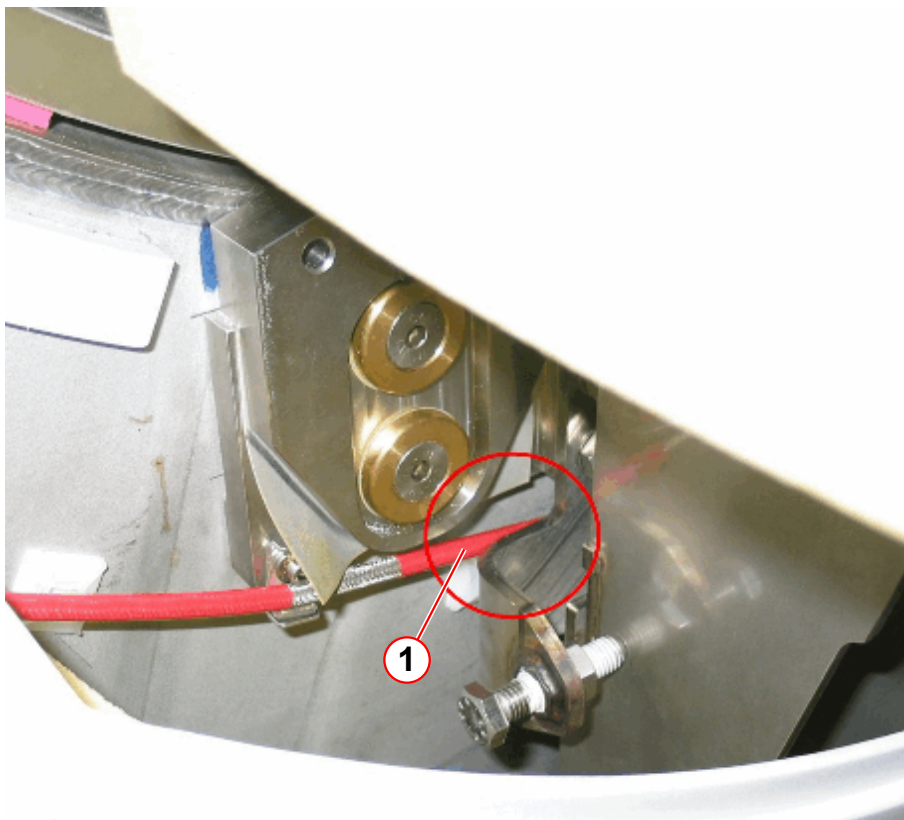


- (1) One grounding point without rear support  
(2) Two grounding points with rear support



For systems with rear support, the RF cable BC\_0 must be fed through the gap between rear support and magnet.

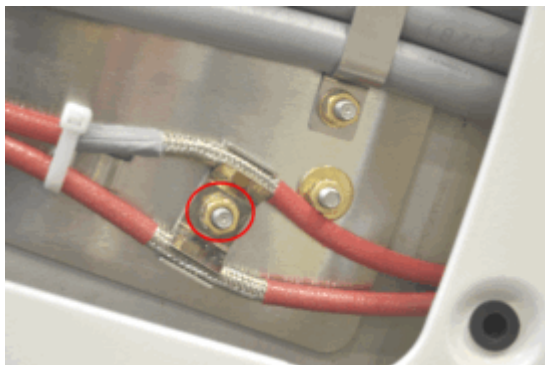
Fig. 263: RF cable routing with rear support



- (1) Feed cable BC\_0 through gap between rear support and magnet

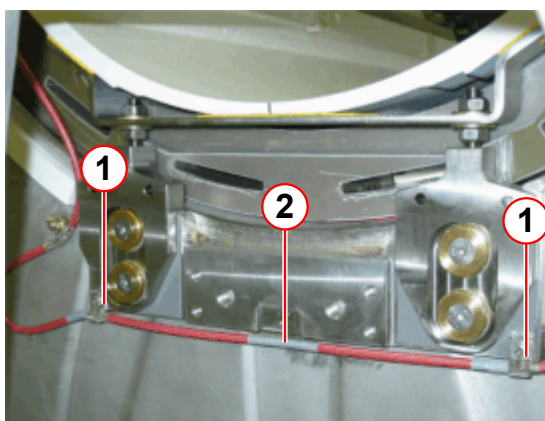
- Tighten the clamps properly to avoid cable movement and spikes.

Fig. 264: Ground connection clamps - RF-cables



To prevent spikes, insulate unused ground shields with tape.

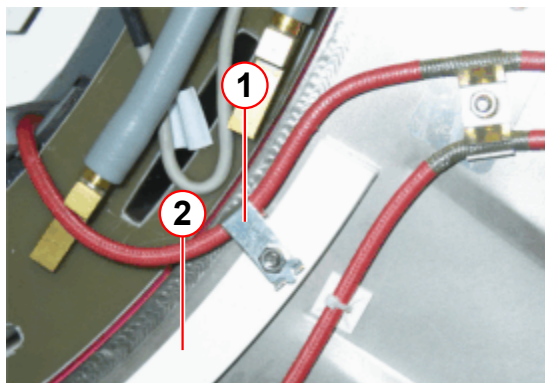
Fig. 265: RF cable insulations



- (1) Used ground point
- (2) Unused ground shield - insulated with adhesive tape

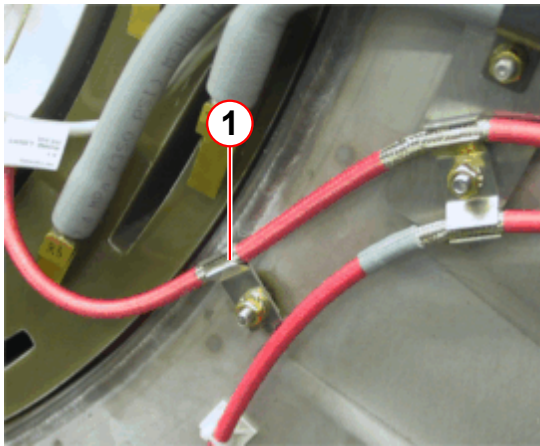
**There are 2 additional RF-cable clamps at newer Aera systems:**

Fig. 266: Additional mounting clamp with Z2-shim



- (1) RF cable mounting clamp
- (2) Z2-shim segment

Fig. 267: Additional mounting clamp without Z2-shim



(1) RF cable mounting clamp

- Reconnect the RF-cables at the TX-Box (BC\_0/BC\_90).

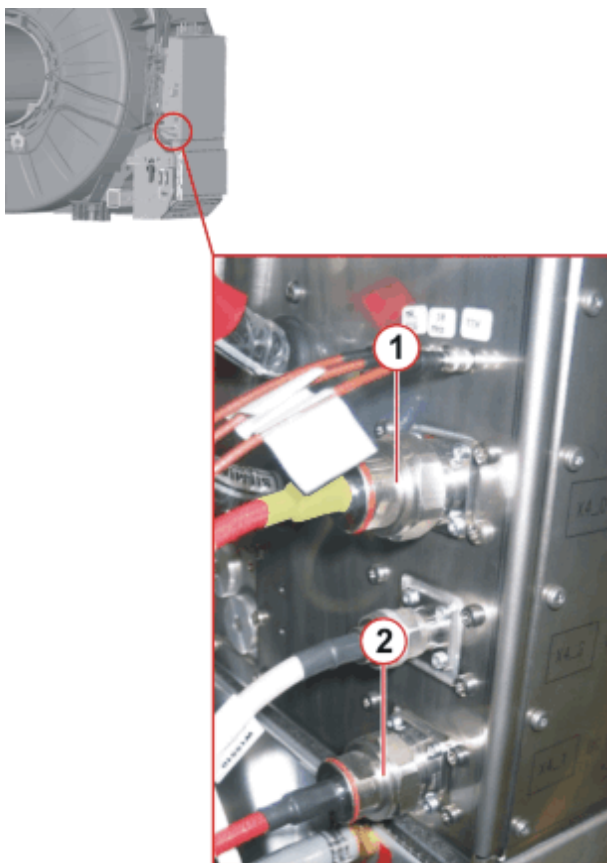


Observe the specified torque for the RF-connectors at the TX-Box (see label at TX-Box):

| TX-Box MR-Frequency | X4_0 BC 0        | X4_1 BC 90       | X4_2 LC         |
|---------------------|------------------|------------------|-----------------|
| 63.6 MHz            | 1.5 Nm ( 1-2 Nm) | 1.5 Nm (1-2 Nm)  | 1.5 Nm (1-2 Nm) |
| 123.2 MHz           | 15 Nm (10-20 Nm) | 15 Nm (10-20 Nm) | 1.5 Nm (1-2 Nm) |



Fig. 268: Transmit cables at TX-Box - Example



- (1) TX-cable BC\_0
- (2) TX-cable BC\_90

#### 23.4.4 Completion of Mechanical Parts

- Re-attach the BC rail parts inside the bore.



- Re-attach the black, sound-absorbing foam into the gap between BC and GC (only at magnet rear-side). Use a flat, non-magnetic tool (e.g. stainless steel or plastic ruler) to fit the foam strip into the gap.

**Foam insertion procedure (magnet rear-side only):**

- a) Press the flat tool against the leading edge of the foam and insert the strip carefully into the gap between GC and BC.

*Fig. 269: Correct fitting - foam inserted at the leading edge first*



*Fig. 270: Incorrect fitting - foam clumped together*



**NOTICE**

Incorrectly fitted foam can lead to image quality problems.

Foam pads which are twisted or clumped together can couple mechanical vibration from GC to BC, leading to signal voids at diffusion weighted images (DTI).

» Insert the foam pads smoothly, do not twist or clump together!

- b) Reposition the flat tool to the middle of the strip and continue inserting the foam.

*Fig. 271: Push the strip inside - tool in middle of the foam*



- c) Reposition the tool to the outer edge of the strip inserting the foam further inside.

*Fig. 272: Push the strip farther in - tool at outer edge of the foam*



- d) Continue until the foam strip fully aligns to the edge of the gradient coil.

*Fig. 273: Foam strip fully inserted*



- e) Continue to insert the foam strip along the circumference of the BC.

*Fig. 274: Inserting the foam along the BC circumference*



- f) Here, at 12 o'clock position, the first strip ends - continue with a new one. Make sure there is no gap left between ascending strips to avoid acoustic leaks.

*Fig. 275: 12 o'clock end of first strip - continue with the next one*



Leave no gap between ascending foam strips to avoid acoustic leaks.

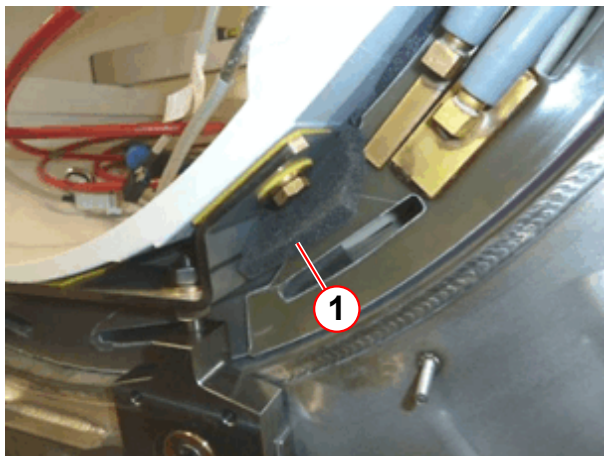
- g) Finish the insertion of foam strips along the whole circumference of the BC.

*Fig. 276: Finish insertion of foam strips*



- h) Cut the last strip to the required length

*Fig. 277: Cut the last foam strip to required length*



- (1) Excessive foam length - needs to be cut

- i) The picture below shows a correctly inserted foam which aligns to the GC edge.

Fig. 278: Finished - correctly inserted foam strip



- Re-install the foam blocks at magnet rear-side and front-side.

Fig. 279: Foam blocks rear-side

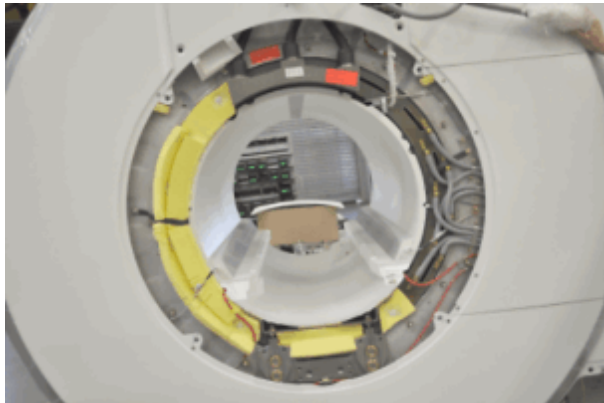
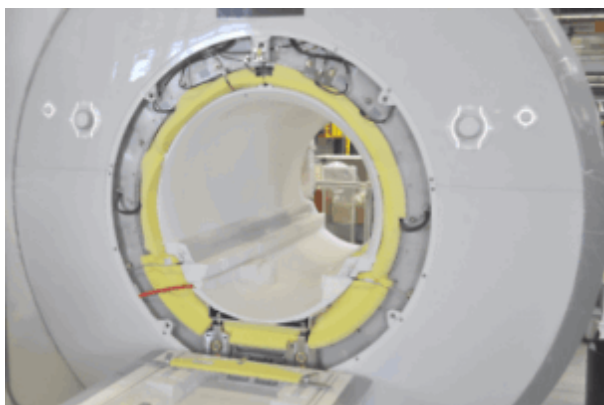


Fig. 280: Foam blocks front-side



- Re-assemble the system covers, except the rear funnel (required for the following BC-Tuning adjustment).



## 23.5 Start-up



The Optical Length Meter (OLM) inside the BC is discontinued. Even, if still present at the old BC it should not be re-installed at the new BC.

Check / perform "Deactivation of OLM" in the following Update Instructions together with a BC replacement:

- MR005/15/P + MR019/15/R (Aera, Skyra)
- MR021/15/P (Skyra pTX)
- MR023/15/R (Aera 24Ch)

Schedule 30 min. additional time for the Update.

### 23.5.1 Mechanical Work Steps

- Switch on the circuit breaker F1/GPA and F21/EPC.
- Switch on the MR system at the alarm box.
- Pack the old BC including all delivered special tools and unused material in the shipping crate. The bundled BC-cables must be attached to the coil with adhesive tape

### 23.5.2 Software



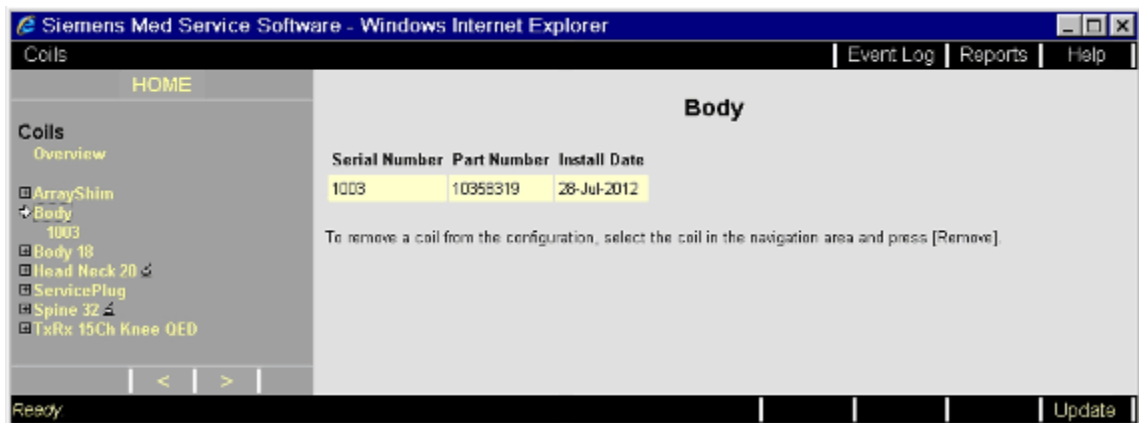
Starting with software version VD13A, the new Body Coil must be registered under **Service Software>Coils**. The old Body Coil serial number must be deleted there.

In case SW VD13A or higher is installed, the new Body Coil must be registered under **Se-So>Coils**, otherwise **NO TuneUp or measurement is possible**:

- Start Service Software
- Start "Coils" menu
- Select **Body Coil>Update**; the new Body Coil should be visible now
- Select the old BC serial number, click **Remove** to delete the old BC



Fig. 281: SeSo&gt;Coils Menu - Example



### 23.5.3 TuneUp/ Adjustment



The "BC Tuning" measurement will be performed with the large spherical phantom. If the "BC Tuning" is out of specification, a manual readjustment of the BC trimmer capacitors is necessary.

- Perform TU/ BC Tuning (refer also to Online Help).
- Perform TU/ BC Power Losses.
- Perform TU/ Image Brightness.

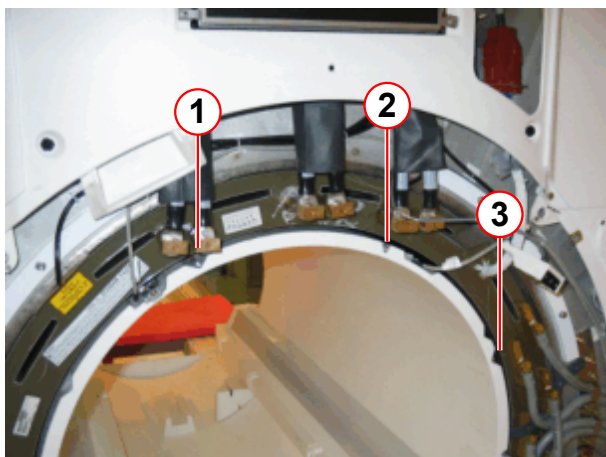
#### 23.5.3.1 BC Tuning

For accessing the trimmer capacitors at the BC rear-side, the rear funnel cover must be removed.



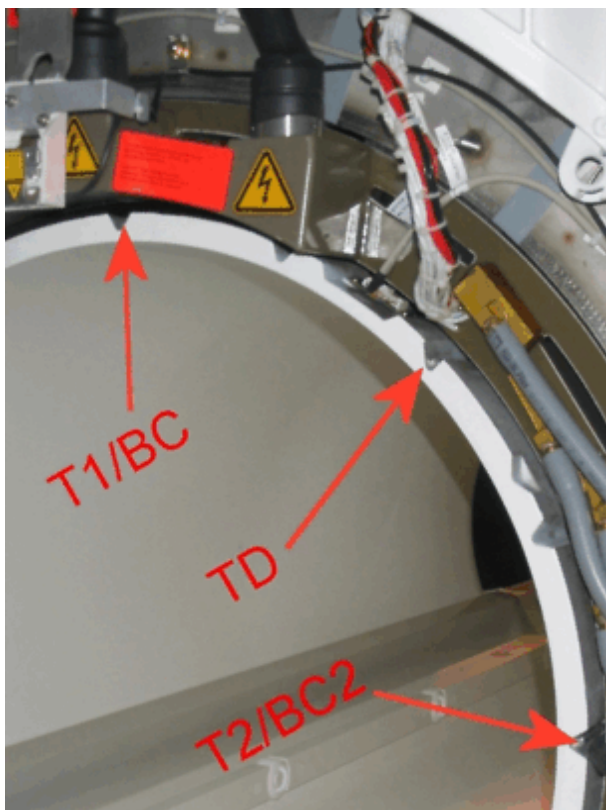
The BC trimmer capacitor positions are slightly different for standard systems or Skyra systems with "Tim TX True Shape" option installed (see following figures).

Fig. 282: BC trimmer capacitors rear-side (standard system)



- (1) Adjustment screw for trimmer T1 (BC2)
- (2) Adjustment screw for trimmer TD (BC, BC2 and Transmission)
- (3) Adjustment screw for trimmer T2 (BC)

Fig. 283: BC trimmer capacitors rear-side (TIM TX True Shape option)



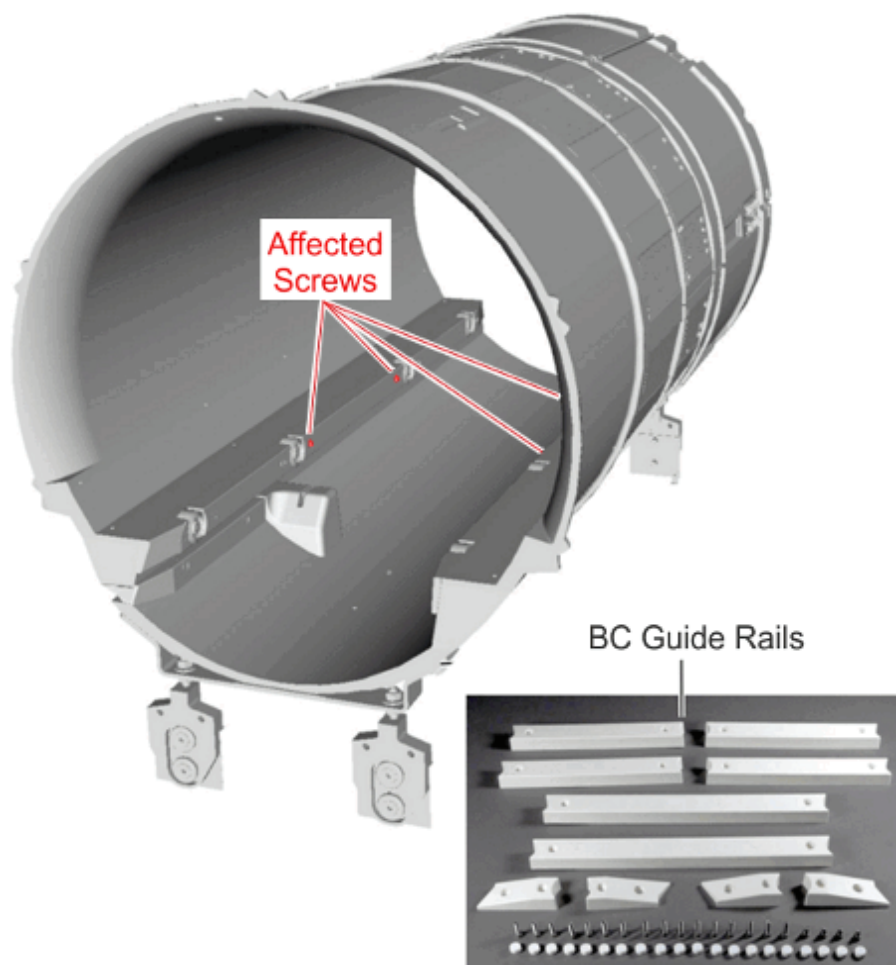
For the BC-Tuning adjust procedure, refer to the **Service Software>Online Help**.

- Re-assemble the rear funnel cover.

### 23.5.4 Function Tests/Q Tests

- Perform QA/ BC Tuning Check.
- Perform QA/ BC Image Check.
- Perform QA/ BC Power Losses Check.
- Perform QA/ Spike Check.
- Only for TimTX True Shape option: Perform QA/ PTX Image Check additionally.
- Perform QA/ Phantom Shim Check
  - Compare the new QA/ Phantom Shim results with the latest valid ones.
  - Compare the field terms, particularly term B(3,1) and term A(4,0).  
If the difference is greater than 1ppm or the QA is out of specification, then perform the following steps (otherwise stop here):
  - Remove the 4 inner guide rail screws from the new BC.
  - Use the slightly magnetic screws from the old BC to reestablish the old shim situation.

Fig. 284: BC guide rail Aera, Skyra



## 23.6 Final Steps

- Ensure that the shim is in specification by running QA/ Phantom Shim Check.